



Department of Computer Science and Engineering

Academic Year 2022 - 2023 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: CS8603 & Distributed Systems

Name of the Faculty member (s): Dr.V.Anusuya, ASCP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit I / **Logical Time: A framework for a system of logical clocks – Scalar time –Vector time**
- **Course Outcome:** The students will be able to elucidate the foundations and issues of distributed systems
- **Topic Learning Outcome:** TLO2
- **Activity Chosen:** Learning by Teaching
- **Justification:**

In unit I, the students need to understand the concept of logical clock, scalar time and vector time. This activity helps the students to understand the logical clocks.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - A topic Logical clock was given to Ms. Dhivya M.K one week before the presentation.
 - She prepared the presentation and taken the class on 22.02.2023
 - The topic was delivered around 30 minutes.
 - The students first understand the need of physical and logical clock, that was taught in the previous class. Divya asked the need of clocks, the students answered the questions.
 - Then she continue with logical clock and explain about Lamport algorithm and casual ordering.
 - The implementation of Learning by teaching activity as shown in figure1 and figure2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO10	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	2	1	3	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO10	PO12	PSO1	PSO2	PSO3
	3	2	1	2	1	3	3	2
Justification for correlation	Apply theory and principles of Computer Science and Engineering to know the concept of logical clocks	The students will be able to analyze the complex problems by distributed computing	Solve the complex problems by distributed computing using various design alternatives	The students can be able to explore the technical knowledge of the model of distributed computing using logical clocks.	The students will be able to explore the historic evolution of distributed computing	As knowledge about the logical clock in distributed system technique used in mobile application development.	The concepts of logical clock can be used in IoT, Cloud computing and cybersecurity	The students exhibit knowledge to solve big data analysis using distributed computing

- Images / Screenshot of the practice:



Figure 1: Learning by Teaching by Divya MK



Figure 2: Learning by Teaching

- Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- All the students were eagerly listened the class. They understand the concepts of scalar time, vector time.
- Students explored the importance of logical clock in distributed computing.
- It makes the students to gain knowledge of Lamport's snapshot algorithm for logical clocks and its synchronization.

❖ ***Benefit of the practice:***

- Students can able to understand the role of Lamports algortithm in logical clock.
- Students can able to know the concepts of Physical and logical clock.
- This activity helps the students to know importance of importance of clock and its synchronization.

❖ ***Challenges faced in implementation:***

- First students difficult to understand the concept. I recalled the concepts of physical clock and the importance of synchronization with an example.
- The topic was planned for 20 minutes, but she takes time around 30 minutes.

References:

- ❖ <https://ohioline.osu.edu/>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/>

Signature of Faculty Member

Dr.V.Anusuya

HOD

Dr.K.Vijayalakshmi